Where is the history effect carried?

A brief summary of the residual-state experiments

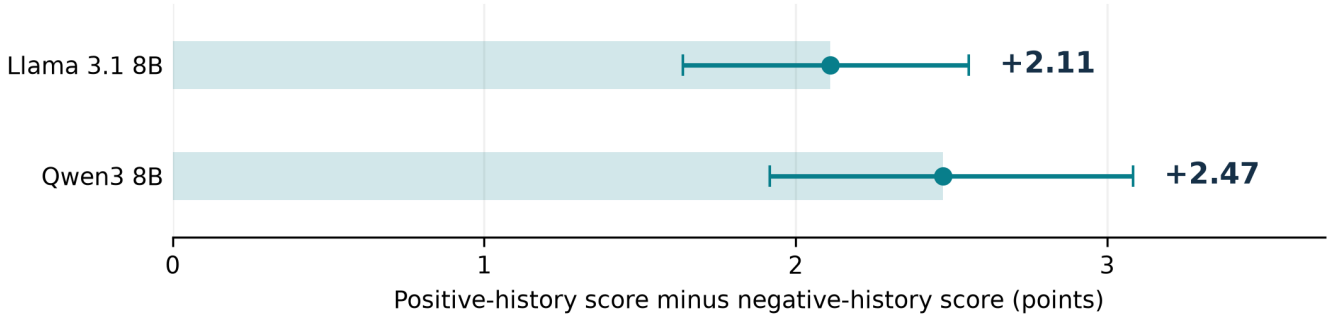
Main finding: positive versus negative history strongly changes scores. Replacing one tested internal state usually does not change the score. We have small leads, but no confirmed mechanism on unseen tasks.

What we did

Llama 3.1 8B and Qwen3 8B judged the same frozen answers under a reported prior rating of 10/10 or 2/10. Each model used the same 18 discovery tasks: six each from MT-Bench, Arena-Hard and WildBench.

We copied internal prompt states between the two histories, in both directions. We tested 24 sites per model: eight layers (4 through 32, every four layers) at three locations: the last history token, the answer tokens, and the final prompt token. The model then generated its own evaluation and numeric score.

History still changes the score



Mean paired score gap on a 0-10 scale. Bars show stratified 95% bootstrap confidence intervals; 18 tasks per model. All 18 tasks showed a positive gap in both models. These are the new discovery tasks, not the earlier 72-task experiment.

Model	10/10 history	2/10 history	Gap
Llama 3.1 8B	9.33	7.22	+2.11
Qwen3 8B	9.69	7.22	+2.47

What this tells us

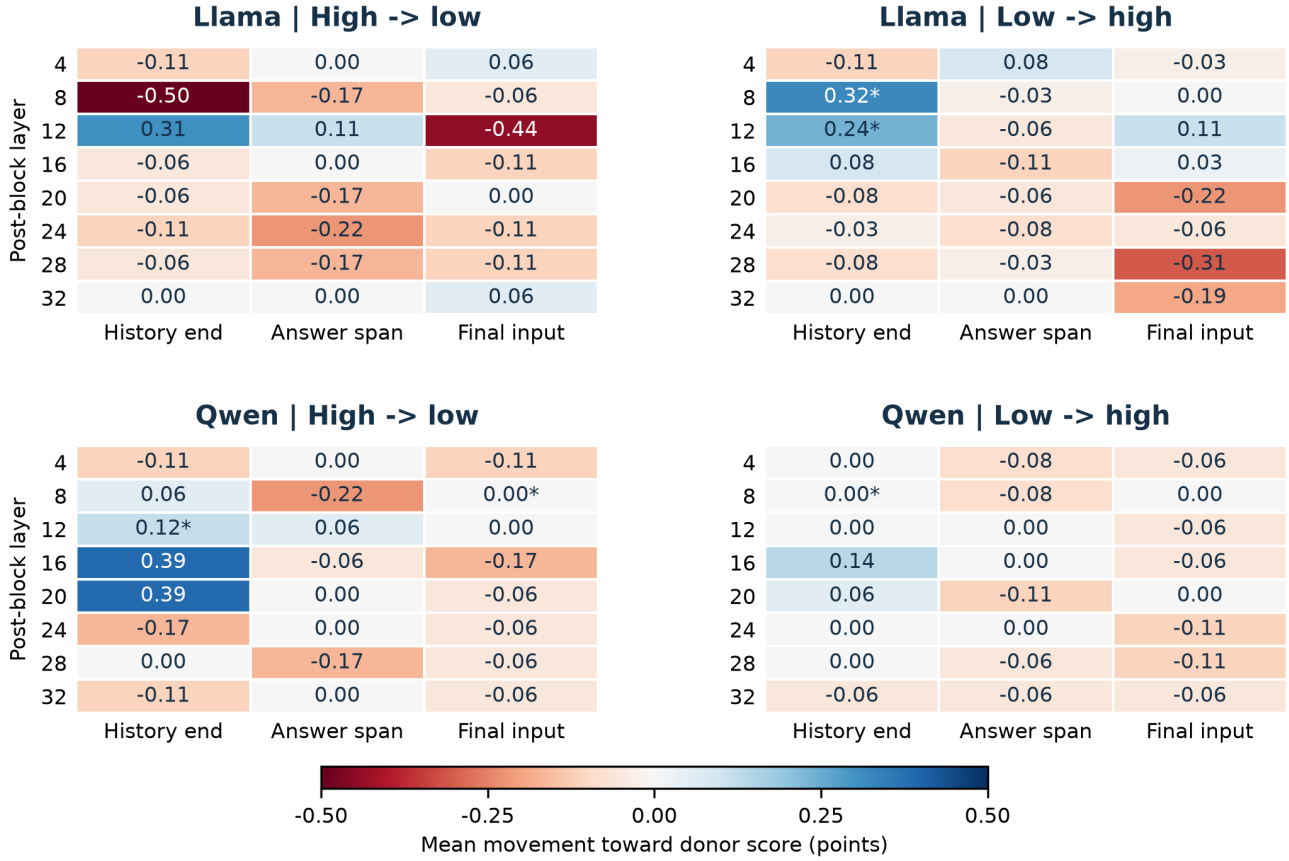
The behavioral effect is clear. The intervention asks a narrower question: can a particular prompt state transfer it? A weak patch does not mean history has no internal effect. Information may be distributed, or the model may read history again while generating.

Did the checks pass?

Yes: all 3,528 planned discovery records are present. All 1,728 same-state identity checks and 24 embedding-recovery positive controls passed. Five outputs were unusable (four hit the token limit and one could not be parsed); they remain missing, not zero. Saved effects were reproduced from the raw records.

WHAT THE INTERNAL INTERVENTIONS SHOW

Internal replacements: small effects



Each cell is the average score movement toward the donor history after replacing one site. Blue means movement in the expected direction; red means the opposite. H to L copies high-history states into the low-history run; L to H reverses this. An asterisk marks a cell with a missing usable score.

No score change in 79.5% of usable Llama patches and 89.4% of usable Qwen patches. This compares each patched score with its matching unpatched baseline.

Exploratory lead	High to low	Low to high
Llama: layer 12, last history token	+0.31 points	+0.24 points
Qwen: layer 16, last history token	+0.39 points	+0.14 points

Leads were chosen after searching the 24 sites, using the smaller of the two directional means. Each direction has 18 usable tasks except Llama low-to-high (17). Median movement is zero in all four lead comparisons. Their exploratory 98.75% intervals reach or cross zero and do not correct for the site search. These are hypotheses, not validated discoveries.

What can we say in the rebuttal?

A broader, controlled residual-state search confirms the behavioral sensitivity but does not establish a strong single-site carrier of the effect during free generation. Our discovery results constrain the mechanism claim; confirmation on jointly audited unseen tasks remains future work.

Next, as separate follow-ups: block history reading only during generated tokens; combine that with prompt-state replacement; then compare normal and fixed-rationale scoring. These experiments have not been run here.